

Matrix inequalities and norms

Thompson-type domination for the arithmetic symmetric modulus

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: The arithmetic modulus lacks the useful order structure of the quadratic modulus; a solution would advance a focused family of operator triangle estimates.

Resolution — 2026-09-11

Negative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS.**

No finite constant permits the proposed two-unitary Loewner-order domination for the arithmetic symmetric modulus, already in dimension three. A fixed rational example also refutes the proposed $\sqrt{2}$ constant. This concerns matrix order, not a separate norm triangle inequality.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Problem statement

For $X \in \mathbb{C}^{n \times n}$, define $|X| = (X^*X)^{1/2}$ and

$$S(X) = \frac{|X| + |X^*|}{2}.$$

For every $n \geq 1$ and every $X, Y \in \mathbb{C}^{n \times n}$, do there exist unitary $U, V \in \mathbb{C}^{n \times n}$ such that

$$S(X + Y) \leq \sqrt{2}(US(X)U^* + VS(Y)V^*)?$$

Here $P \leq Q$ means $Q - P$ is positive semidefinite. The conjectured universal factor $\sqrt{2}$ is already known to be necessary.

Why it matters

Symmetrizing the left and right polar factors removes their directional preference. The question asks for a precise matrix-order bound for sums, which would yield singular-value and norm consequences for these positive representatives.

References

1. T. Zhang, *Operator symmetric moduli and sharp triangle inequalities*, arXiv:2603.01046v1 (1 March 2026), §1.2, Conjecture 1.6. [Primary text](#).
2. J.-C. Bourin and E.-Y. Lee, *Some hybrid matrix triangle inequalities*, arXiv:2606.29188v1 (28 June 2026), introduction, Theorems 1.1 and 1.3. [Primary text](#).

Status check — 2026-09-10

Both latest arXiv records remain v1. The June paper supplies related estimates but no proof of the displayed two-unitary inequality. Searches included *Zhang Conjecture 1.6 symmetric moduli*, *arithmetic symmetric modulus Thompson conjecture 2026*, and the two titles with *proof*. No resolution was located. The analogous theorem for the quadratic symmetric modulus is proved and must not be confused with this conjecture; a formula labeled Conjecture 3.13 in arXiv:2606.15624 appears to misidentify the modulus, so the draft follows Zhang’s original statement.

Audit update (2026-09-10): Rechecked Zhang Conjecture 1.6 and Bourin–Lee Theorem 1.3, with searches for later symmetric-modulus resolutions. Submajorization is weaker than the required two-unitary positive-semidefinite domination. This is a bounded literature check, not a proof that no solution exists.